Entrepreneurial Ecosystem Emergence

Overview of the Theory, v1.72 Draft

Status of this draft

This overview presents the theory of entrepreneurial ecosystem emergence at the mean-field architecture level. Version 1.72 conforms the overview to the present state of the theory-development record. The conceptual definition of participation is consolidated under recently ratified rulings; the relationship between local and aggregate structural description is articulated through the distinction between geographic space and the space of structural conditions; the orientation feature of action streams is given an explicit structural guard; the state-dependence of interventions is stated; and a forward articulation concerning the sign of coherence is planted for subsequent development. These refinements improve how the theory is explained and record recent consolidations. They do not alter the core architecture.

Computational substrate work now underway extends beyond this overview. That work tests candidate formulations and substrate implementations of open elements in the theory. The overview below should therefore be read as a statement of the theory at the mean-field approximation (MFA) level, with the computational substrate program continuing as the next stage of formal exploration. (The substrate program realizes the theory at agent grain using stochastic lattice methods; it is loosely describable as agent-based modeling, though it inverts that tradition's usual mode: the theory lives at the collective level, and the agents deliberately carry no behavioral hypothesis.) This overview is intended to add clarity and color to the austere mathematical objects that explain the mechanics of entrepreneurial ecosystem emergence.

Many of the assertions in this overview — about action streams, orientation, structural commitments, and the level at which the theory operates — rest on lineages of prior scholarship that manuscript-stage work will need to engage formally. Defensibility against the relevant fields' scholarship is manuscript-stage work and is not undertaken here.

1. Why a theory of entrepreneurial ecosystem emergence is needed

Entrepreneurial ecosystems are often described through visible ingredients: entrepreneurs, startups, investors, mentors, universities, accelerators, policies, events, networks, and culture.

These ingredients matter. They are part of the world practitioners recognize. Yet the presence of ingredients for an entrepreneurial ecosystem does not by itself explain the emergence of one.

Some communities contain founders, programs, events, capital sources, and support organizations without developing the character of an entrepreneurial ecosystem. Activity occurs, but it remains fragmented. Founders may build companies, serve customers, seek financing, hire employees, and solve hard problems while still experiencing entrepreneurship as mostly solitary work, concerned chiefly with matters concerning their own firms.

Other communities acquire a different character. Entrepreneurs begin finding one another. Their problems become mutually recognizable. They see that founders share the same problems even though their businesses don't have much in common. People benefit from the relationships and introductions compound. Mentors appear because prior founders remain engaged. Stories of entrepreneurial effort become more visible and legitimate. Participation begins to reinforce future participation.

The theory asks why entrepreneurial activity in some communities remains isolated or episodic, while in others it becomes connected, persistent, and mutually reinforcing. The theory approaches the question structurally. It treats entrepreneurial ecosystems as emergent systems arising from many action streams unfolding under particular structural conditions. The aim is to explain how entrepreneurial action becomes organized into a collective structure, and how that collective structure comes to reinforce the conditions of further entrepreneurial participation.

2. The phenomenon to be explained

Entrepreneurial activity is not rare in the ordinary sense. People start businesses, develop products, serve customers, file taxes, hire workers, look for capital, and struggle through the practical demands of building companies in many kinds of places. That activity matters, but it does not always amount to ecosystem emergence.

A founder may work intensely with a small team, a set of customers, suppliers, regulators, accountants, lawyers, markets, and banks. The activity is entrepreneurial, but it can remain structurally solitary. The founder is doing the work of entrepreneurship without necessarily participating in a broader entrepreneurial community.

A different condition appears when entrepreneurs begin to find the company of others going through analogous struggles in their own contexts. They hear stories that make their own experience more intelligible and less uncertain. They meet people who understand the problems they face. They discover mentors, peers, investors, service providers, and institutional actors whose actions alter the conditions under which entrepreneurial work proceeds.

The difference is not simply that there are more entrepreneurs. The difference is that entrepreneurial action begins to acquire relational structure. People begin talking to one another and find unexpected value in the conversations. Information begins to travel through entrepreneurial channels. Introductions begin to matter beyond a single exchange. The local act becomes part of a larger pattern.

This is the phenomenon the theory seeks to explain. Entrepreneurial ecosystem emergence is the process through which entrepreneurial action streams become sufficiently connected and persistent to form an activated domain, and through which that domain becomes mutually reinforcing at the system level.

3. The basic proposition

The theory proposes that entrepreneurial ecosystem emergence occurs through a two-stage cascade. By cascade we mean an ordered dynamic structure in which one transition's outcome becomes the precondition for the next; the formal treatment of this dependency is developed in later sections.

The first transition is from isolated entrepreneurial activity to ecosystem-relevant activation. This does not mean that entrepreneurial activity suddenly appears where none existed. It means that entrepreneurial action begins to acquire a different structural character. Action streams become sufficiently connected and structurally coupled that they form a participating activation domain.

In short, people start paying attention to what other entrepreneurially-minded people are doing. People and organizations perceive collaboration and participation as a benefit. The theory's mechanics do not draw on behavioral or psychological factors. What appears phenomenologically as paying attention is, at the level the theory works, action stream orientation toward an emerging entrepreneurial milieu —

orientation being structured by configurational conditions. The phenomenological surface and the architectural level describe the same phenomenon at different levels of description; the theory works at the architectural level. People are, however, people, and the theory in no way denies these human attributes. These properties do not enter the theory directly because they belong to a different level of description than ecosystem emergence. The theory is only concerned with actions taken over time and the emergence of entrepreneurial ecosystems.

The second transition toward ecosystem emergence is when it moves from activation to coherence. Activated entrepreneurial activity may remain fragmented. Coherence appears when activated entrepreneurial activity becomes mutually reinforcing at the system level. Prior entrepreneurial action alters the conditions of future entrepreneurial action. Participation begins to generate more participation.

The theory therefore identifies three regimes:

- **Inactive regime:** entrepreneurial activation is low, unstable, or structurally isolated.
- **Activated but incoherent regime:** entrepreneurial activity is present, persistent, and increasingly connected, but participation has not yet become coherently self-reinforcing.
- **Coherent ecosystem regime:** entrepreneurial activity becomes mutually reinforcing in ways that reproduce the conditions supporting future participation.

The activated but incoherent regime is central to the theory. It explains why some communities are busy but not coherent. They may contain programs, events, founders, and energy without producing the mutual reinforcement associated with ecosystem emergence.

4. Why the theory begins from action

The theory begins from action rather than motivation, preference, intention, or decision. Entrepreneurs are purposeful people. They think strategically, evaluate opportunities, imagine futures, and pursue goals. Those realities remain fully compatible with the theory.

The point of departure is action because the object of explanation is ecosystem emergence. Entrepreneurial ecosystems are collective phenomena. They arise from many entrepreneurial actions interacting under structural conditions over time. Action is the point at which individual conduct enters collective dynamics.

This is a claim about the level at which the theory operates. The motivators that drive any particular agent are not the same object of study as the structural conditions that govern whether an ecosystem forms. These are different phenomena at different levels, and they require different modes of description. A theory of ecosystem emergence is not a theory of individual behavior in compressed form, and should not be read as one.

Action also provides empirical grounding. Actions can be observed and aggregated. Internal states can be inferred only indirectly. But this is the secondary reason. The primary reason is that the theory describes a level of phenomena at which interior states are not the appropriate variables.

The primitive of the theory is therefore the entrepreneurial action stream. An action stream is not a single act. It is a continuing pattern of entrepreneurial doing: forming a company, serving customers, seeking financing, hiring employees, building products, attending events, giving advice, introducing

people, mentoring founders, joining founder conversations, and taking other actions that enter entrepreneurial dynamics.

The action-stream primitive carries a structural commitment worth making explicit. A constitutive feature of action streams is their orientation toward some structural conditions and away from others. The theory's refusal of optimization, utility, and decision rests on this commitment. Optimization and utility require orientation to be computed from preferences, goals, or values held by the agent. The action-stream primitive does not derive orientation this way. Orientation is the floor of the architecture, not an output of the architecture. This is what makes the theory's treatment of agents coherent rather than austere. Agents are not stripped-down decision-makers; they are a different kind of primitive entirely, where the question of what orients action is answered structurally rather than psychologically.

Two guards keep this articulation aligned with the rest of the architecture. First, orientation carries no valence. “Toward” and “away” describe the structural directionality of a stream's registration, not attraction, aversion, appetite, or affect; nothing in the mechanism evaluates conditions as good or bad. Second, orientation is not a hidden disposition doing explanatory work behind the scenes. What orientation amounts to, operationally, is stated in Section 5: the terrain determines the region of projection attainable from a situation, and orientation is realized — or not — as projection within that region. Orientation and projection are one articulation seen from the stream's side and from the conditions' side, respectively.

The theory asks when such action streams become structurally relevant to ecosystem emergence and when their collective interaction produces system-level organization.

5. From life activity to entrepreneurial action streams

Human life activity is vastly richer than the theory represents. People act across family, work, civic life, consumption, friendship, religion, politics, health, education, and ordinary daily living.

The theory performs a structural compression. This compression retains only those dimensions necessary to explain how entrepreneurial action becomes persistent, connected, and collectively organized. The theory is therefore not reducing people to variables. It is retaining a small set of structural dimensions relevant to entrepreneurial ecosystem emergence.

Participation enters at this point. Participation means that an action stream has become relevant to the entrepreneurial dynamics the theory is modeling. The term is used structurally, not as identity, permission, or formal membership.

Some action streams project onto the structural basis system of the theory; many do not. Projection here names the meeting between an action stream's orientation and the structural conditions that support or inhibit it. An action stream that projects has found terrain — viability, transition cost, social reinforcement — on which its orientation operates. A person's life contains far more action than the theory describes. The theory retains only those entrepreneurial-relevant action streams that can enter the cascade from activation to coherence.

The conceptual definition of participation has now been consolidated. An action stream participates when, over a window of time, it registers non-zero on viability and on social reinforcement, with transition cost below its prohibitive limit, all evaluated at the agent's own situational configuration of the bases. All three conditions must hold together: if any base sits at its annihilating extreme, the stream is invisible to the theory's aggregate variables regardless of the other two. The window is an

architectural parameter — a declared timescale on which participation is defined — rather than a tuning choice. Participation is the primitive, and activation density is derived from it: the density aggregates registrations; it does not define them.

Two distinctions do real work here. An agent can be acting — visibly, energetically, even entrepreneurially in the everyday sense — while failing to register on some base; such an agent is acting but not participating. And favorable conditions permit participation without producing it: the theory registers that a stream did not project, and requires no account of why. What remains open is not the definition but its empirical operationalization: the formal rule by which particular observed action streams are identified as registering, using data that was collected under other descriptions. That identification problem is an instance of a broader measurement question the research program has named and reserved for dedicated work.

One boundary of the definition deserves explicit note. Streams that register on some bases but not all are real and describable — they are part of the texture of a place — but they do not constitute a second population with its own theoretical standing. The aggregate variables are defined over participating streams within the relevant domain; partial projection is heterogeneity within the picture, not a constituency beside it.

This matters because the aggregate variables of the theory are meaningful only over the domain being described. Activation density, for example, is not density over all people or all life activity. It is density over the relevant domain of entrepreneurial action streams.

6. The structural bases: v, c, and r

The state space the theory works within has three structural bases:

- **viability (v),**
- **transition cost (c),**
- **social reinforcement (r).**

These bases describe structural conditions under which entrepreneurial action unfolds. They are the dimensions of the participation space.

Viability refers to whether entrepreneurial action is materially possible under present conditions. It includes conditions such as time, health, resources, capability, legal standing, and related enabling factors.

Transition cost refers to the friction involved in moving into entrepreneurial participation. It includes forgone wages, lost benefits, regulatory burden, uncertainty, sunk costs, reputational exposure, and other barriers that make entrepreneurial action costly to undertake.

Social reinforcement refers to the degree to which entrepreneurial action is supported, normalized, and reinforced by the surrounding social environment. It includes mentors, peer examples, family attitudes, visible successes, narratives of legitimacy, and network encouragement.

Four points about the bases are important to keep in view together. The bases describe the structural conditions under which an agent acts; they are not variables interior to the agent. Their values are agent-specific: two entrepreneurs in the same city may face very different configurations of viability, transition cost, and social reinforcement. The bases are independent dimensions of the participation

space, not weighted contributions to a single quantity. And while their values change over time, they change in response to the system's own activity through the feedback operator Q, introduced in Section 9.

7. Two spaces, local conditions, and aggregate variables

A central clarification concerns two different senses of “space” the theory uses, and the relationship between local structural conditions and aggregate collective variables.

Geographic space is where agents are located: neighborhoods, districts, regions. The space of structural conditions is different: it is the three-dimensional space whose coordinates are viability, transition cost, and social reinforcement, in which each point is a possible configuration of conditions. The terrain of a place is the map between the two — the assignment of structural coordinates to geographic locations. Because many agents occupy any real location, and the conditions bearing on each are situational, a location is best pictured as a distribution of positions in the space of structural conditions rather than as a single point. Two entrepreneurs in the same city — even in the same building — may occupy very different positions.

Λ is defined on the space of structural conditions: it is the composition of the bases into a single control quantity, and it can therefore be evaluated anywhere in that space. Evaluated at an agent's situational configuration, Λ is local: it is the one channel through which structural conditions shape that agent's action stream. Coarse-grained across the many situated streams of a region, Λ is the theorist's aggregate descriptor of the structural conditions those streams collectively encounter. The local and the aggregate readings are the same function at two resolutions, and keeping both in view prevents two opposite mistakes.

The first mistake is reifying the aggregate. Agents do not respond to the aggregate Λ as a stimulus or assess it as a landscape; no agent perceives the region's conditions, and each acts under its own. The second mistake is severing the local. The theory does commit that structural conditions reach agents — through the locally evaluated control quantity and through nothing else. Agents have no additional channel by which the macro state addresses them directly. What appears at the system level as a response to aggregate conditions is the statistical composition of many streams, each shaped only by its own situation.

The architecture moves through levels:

- broader life activity,
- entrepreneurial-relevant action streams,
- projection onto retained structural bases (v, c, r),
- coarse-graining into aggregate descriptors such as Λ,
- system-level dynamics of activation (ρ) and coherence (Ψ).

This layered structure allows the theory to connect lived entrepreneurial action to formal collective dynamics without collapsing one into the other.

8. Independence of the bases

The retained bases are independent in two senses that must be kept distinct.

They are independent in their values at any given moment. High social reinforcement does not make action possible when viability collapses. Low transition cost does not compensate for the complete absence of viability. High viability does not by itself create social reinforcement. The theory captures this through a zero-collapse property: if any necessary structural base reaches its limiting boundary, the control quantity Λ collapses regardless of the values of the other bases. The bases do not function as a weighted index. Entrepreneurial participation depends on a joint structural configuration in which each base contributes something the others cannot.

They are also independent as dimensions of the participation space. Each base captures a structurally distinct condition for action: viability captures whether action is materially possible, transition cost captures the friction of moving into participation, and social reinforcement captures whether the surrounding social environment supports the action. These are not three facets of a single underlying quantity. They are three non-substitutable structural conditions, and the theory commits to their dimensional independence as a constitutive claim about what an entrepreneurial ecosystem is.

This dimensional independence is preserved through time by the architecture of Q, the feedback operator introduced in the next section. Q acts on the bases individually rather than on the aggregate Λ. The reason is precisely to preserve the differential structure of the state space: if Q acted on Λ directly, the bases would lose their identity as distinct dimensions and the architecture would collapse to a single-variable description.

The exact function linking (v, c, r) to Λ remains an open element at this level of the theory. The architecture commits to the bases, to their independence at any moment, and to their dimensional distinctness through time, not to a final closed functional form.

9. Historical evolution and Q

The bases are independent at a given moment, but they are not fixed over time. Entrepreneurial ecosystems evolve historically. Activity in the system can reshape the structural conditions under which future entrepreneurial action occurs. This is the role of Q, the feedback operator.

Q describes how the macro-state of the system feeds back into the structural bases over time. Sustained entrepreneurial activity can alter viability, transition costs, and social reinforcement. It may create resources, normalize entrepreneurial participation, thicken networks, build institutional routines, lower barriers, or make entrepreneurial roles more available.

A simple practical example is instructive. Entrepreneurs who value their startup community may talk about it, mentor newer founders, take meetings when asked, make introductions, and publicly legitimate entrepreneurial effort. These actions strengthen the conditions under which future entrepreneurial participation becomes more likely. They reinforce participation through speech and action.

A further regularity belongs here because it follows from the architecture rather than from any property of actors. The effect of an act on the system is a joint property of the act and of the system's state when the act lands. Near a transition, the system's responsiveness grows; far from one, it shrinks. The identical intervention — the same program, the same speech, the same fund — can be inert in one condition of the system and decisive in another. This is the theory's explanation for two stubborn practitioner observations: intervention playbooks that transform one community and do nothing in another, and community folklore that credits particular ordinary moments with extraordinary

consequence. The act was ordinary; the system was near a threshold, and some small perturbation was going to be the one that mattered.

Importantly, Q acts on the bases rather than directly on Λ. This preserves the distinctness of viability, transition cost, and social reinforcement as dimensions of the state space, with the consequence that the bases remain separately identifiable and separately reasonable about across the system's history. If Q acted on Λ directly, the three-dimensional state space would collapse to a single evolving scalar, and the theory's account of why the bases matter would lose its grounding.

The theory therefore distinguishes two forms of relationship among the bases. At any given moment, the bases remain independent. One does not substitute for another. Over time, the bases may become historically correlated because system activity reshapes them through Q.

This is where path dependence enters. The system remembers because prior entrepreneurial activity alters the conditions under which future entrepreneurial action unfolds.

10. The first transition: ecosystem-relevant activation

The first transition is best understood as a change in the structural condition of entrepreneurial activity.

Entrepreneurial action can exist in a community while remaining solitary or weakly coupled. Founders may be building companies, solving customer problems, and navigating markets without their actions becoming part of a broader entrepreneurial milieu. The first transition occurs when entrepreneurial action streams become sufficiently connected and structurally coupled that they begin to form an ecosystem-relevant activation domain.

This transition is represented by ρ, activation density. ρ is not simply the number of entrepreneurs or startups. It describes the density of entrepreneurial-relevant activation within the participating domain.

At low values, entrepreneurial action remains isolated and episodic. As structural conditions improve, action streams become more persistent and connected. The participating domain thickens. Entrepreneurial activity begins to acquire an orientation toward an emerging entrepreneurial milieu.

The theory represents this first transition with a Landau-style normal form:

$$d\rho/dt = \alpha(\Lambda - \Lambda^*)\rho - \beta\rho^3 + \eta(t)$$

Activation grows or decays depending on whether the control quantity Λ exceeds a critical threshold. The last term means activation saturates rather than growing without bound.

Λ names the critical boundary at which the inactive regime loses stability. When a regime is stable, small perturbations die out quickly. Λ is not an additional assumption. It follows from the same commitments that produced the equation. If the inactive state were stable at every value of Λ, no ecosystem could ever form: the system would remain inactive regardless of conditions. If the inactive state were unstable at every value of Λ, there would be no genuine inactive state to begin with, and the theory's account of communities that have not yet ignited would have no basis in lived experience and there would be no calls for a theory explaining it.

The theory commits to the existence of Λ^* but not to its specific value. The value depends on the functional form relating the bases to Λ and on the dynamics of Q. It is empirical, contingent, and possibly

history-dependent. What is universal across all ecosystems is that some such boundary exists, and that communities on opposite sides of it are categorically different kinds of systems.

11. Regime II: activated but incoherent

The first transition produces activation, not necessarily coherence. This distinction is central.

A community may contain entrepreneurial density, events, programs, conversations, founders, investors, and visible activity while still lacking coherent ecosystem organization. The theory identifies this as Regime II: activated but incoherent.

Regime II is a distinct structural condition. This regime explains a common practitioner observation: more activity does not always produce coherent ecosystem structure. A community can have many initiatives and still lack the mutual reinforcement needed for coherence.

The theory therefore separates activation from coherence. Activation concerns whether entrepreneurial action becomes dense, persistent, and structurally connected enough to form an ecosystem-relevant domain. Coherence concerns whether that activated domain becomes mutually reinforcing in ways that reproduce the conditions of future participation.

12. The second transition: coherence

The second transition concerns Ψ, coherence. Coherence refers to the degree to which entrepreneurial activity becomes mutually reinforcing at the system level.

In a coherent entrepreneurial ecosystem, individual entrepreneurial actions no longer remain merely adjacent or connected. They begin to reinforce one another in ways that reproduce the structural conditions supporting future entrepreneurial action.

Practitioners often recognize coherence phenomenologically before they can define it formally: information moves faster, introductions compound, trust forms more readily, opportunities travel through networks, experienced entrepreneurs support newer founders, capital finds projects more efficiently, and entrepreneurial activity begins to generate the conditions for more entrepreneurial activity.

The transition into coherence is represented through a second Landau-style dynamic:

$$d\Psi/dt = \mu(\rho)\Psi - \gamma\Psi^3$$

The key term is μ(ρ), which governs whether coherence grows or decays as activation changes. μ(ρ) plays the same architectural role in the coherence equation that $\alpha(\Lambda - \Lambda$*) plays in the activation equation: it is the coefficient whose sign determines whether the order parameter grows or returns to zero. Its dependence on ρ encodes the central claim that coherence requires sufficient activation to become self-sustaining. Below some level of activation, μ(ρ) is negative and any coherence that arises decays back to zero. Above that level, μ(ρ) becomes positive and coherence grows until the cubic term saturates it. The value of ρ at which μ changes sign is the architectural analogue of* Λ in the activation equation.

The theory commits to the distinction between activation and coherence, and to the sign-change structure of μ(ρ). The final closed form of μ(ρ) remains open at the MFA level. That openness preserves

the distinction between the committed cascade architecture and later work testing which functional forms best reproduce the predicted phenomenology.

A forward articulation, planted here for later development: coherence has an orientation as well as a magnitude. The formal structure admits two coherent configurations — mirror images of one another — and in operational terms these read as coherent growth and coherent contraction of participation. The striking consequence is that coherent decline is not a return to the activated-but-incoherent regime: a community can unwind in an organized way, everything moving together, downward. Disorder and decline are different axes. The feedback operator breaks the mirror symmetry over time — coherent contraction writes unfavorable conditions back into the bases and thereby extinguishes its own support — so long-lived ecosystems are growth-oriented not because the contracting configuration is impossible but because it is self-limiting, observable mainly as an episode. This articulation carries the status of a developed conjecture within the modeling program, not a settled element of the architecture.

13. The three regimes

Regime I: inactive

Entrepreneurial activation is low or unstable. Action may occur, but it does not become structurally connected enough to form a persistent activation domain.

Regime II: activated but incoherent

Entrepreneurial activity is present, persistent, and increasingly connected, but it does not yet become mutually reinforcing at the system level. This regime is empirically important because it describes communities with visible entrepreneurial activity but without coherent ecosystem emergence.

Regime III: coherent ecosystem

Entrepreneurial activity becomes mutually reinforcing. Action streams increasingly reproduce the conditions that support future entrepreneurial participation.

The theory's central prediction is that Regime II is real and persistent, not merely a temporary point along a smooth path to coherence.

14. Why the mathematics follows from the ontology

The mathematical form follows from the ontology of the theory. To see why, it helps to be clear about what kind of mathematics is being used.

The theory uses Landau-style normal forms. These are often associated with physics, because that is where they were first developed and most famously applied. But Landau's normal form theorem is not a physical theory. It does not deal with particles, organisms, people, social systems, solar systems, or galaxies. It deals with systems — with the variables that describe them and the symmetry properties of the equations governing those variables. It is a mathematical result about how dynamical systems behave near critical points, indifferent to what the system is made of. The theorem does not know or care whether the components are water molecules, neurons, firms, or entrepreneurs. What it cares about is the structure of the variables and the symmetries of their dynamics.

The theory therefore does not borrow metaphors from physics. It claims something more specific: the variables identified, the commitments made about their behavior, and the symmetry structure of the

system satisfy the conditions under which the theorem applies. When those conditions are met, the form of the dynamics follows.

The theory begins from action streams rather than decisions. It retains structural bases rather than psychological traits. It coarse-grains local conditions into collective variables. It distinguishes activation from coherence. It treats transitions as qualitative reorganizations of system behavior.

These commitments imply dynamics with:

- genuine resting states,
- threshold-dependent growth,
- saturation,
- fluctuations near threshold,
- and separate order parameters for activation and coherence.

Landau-style normal forms are the minimal mathematical structure for representing such regime transitions. The mathematics is used because the theory's commitments require a language capable of describing qualitative changes in collective organization, and because the conditions under which that language applies are the conditions the theory has already specified.

15. Supercritical transitions and the location of path dependence

The theory uses supercritical transition forms at the MFA level. This choice clarifies where path dependence resides.

A subcritical transition embeds hysteresis and bistability directly inside the bifurcation geometry. The present theory instead locates historical memory in Q, the feedback process through which prior system activity reshapes the structural bases.

This separation matters. The cascade describes how activation and coherence respond to current structural conditions. Q describes how the system's history alters those conditions over time. Path dependence therefore lives in the evolving structural conditions, not in duplicated irreversibility inside the bifurcation geometry itself. This keeps the theory cleaner and makes the sources of persistence more distinguishable.

One further layer of the theory's epistemic structure is worth recording. The supercritical form is a committed working hypothesis: it is the specific mathematical housing the modeling program is built to adjudicate, and it could fail while the architecture's deeper commitments survive. Beneath the formalization lies an older and more general claim — that coherence in entrepreneurial ecosystems is real, emergent, and structurally determined. A negative result against the specific transition forms would send that deeper claim in search of different mathematics; it would not refute it. The formalization is the theory's testable surface, not its foundation.

16. Scope of the MFA theory and the role of the substrate program

This overview describes the theory at the MFA level. The MFA architecture establishes:

- the action primitive,
- the retained structural bases,

- the control parameter Λ, read locally at situations and aggregately at the system level,
- activation density ρ,
- coherence Ψ,
- the two-stage cascade,
- and the three-regime structure.

Several elements remain open at this level:

- the final functional form linking (v, c, r) to Λ,
- the detailed form of μ(ρ),
- the empirical operationalization of participation and projection,
- the full structure of Q,
- and the spatial organization of emergence — how coherence first arises somewhere rather than everywhere, and how far it spreads.

The computational substrate program explores these open elements. It realizes the theory at agent grain — many locally situated action streams under locally configured conditions — and tests whether the mean-field picture survives that finite, fluctuating realization, alongside candidate functional forms and operationalizations. It does not alter the core theory stated here. The MFA theory establishes the architecture. The substrate program explores its admissible implementations and adjudicates its committed working hypotheses.

17. Why this theory matters

The theory matters because it explains why entrepreneurial activity alone is not enough. A community may have founders, startups, programs, capital, mentors, events, and institutional support while still failing to produce coherent ecosystem structure.

The theory explains this by distinguishing:

- activity from activation,
- activation from coherence,
- local action from aggregate structure,
- and temporary intervention from historical terrain evolution.

It offers a formal way to understand a practitioner reality often described informally: some communities are busy but not coherent; others reach a point where entrepreneurial activity begins reinforcing future participation. The theory gives language and structure to that difference. It explains ecosystem emergence as a process through which entrepreneurial action streams, unfolding under structural conditions, become connected, activated, coherent, and historically participation-reinforcing. That is the phenomenon the theory is designed to make intelligible.

18. Summary

Entrepreneurial ecosystem emergence is a structural process.

It begins with entrepreneurial action streams unfolding under local conditions. Those conditions are described through retained structural bases: viability, transition cost, and social reinforcement.

The theory compresses these local conditions into aggregate collective variables capable of describing macroscopic system behavior. Λ represents the control quantity composed from the structural bases, read locally at situations and aggregately at the system level. ρ represents activation density within the participating domain. Ψ represents coherence.

The theory proposes a two-stage cascade:

$$(v, c, r) \rightarrow \Lambda \rightarrow \rho \rightarrow \Psi$$

The first transition concerns the movement from solitary entrepreneurial activity to persistent ecosystem-relevant activation. The second transition concerns the movement from activation to coherent, mutually reinforcing participation. Path dependence enters through Q, which reshapes the retained structural bases over time.

The core architecture remains unchanged. Recent refinements clarify how to describe its levels, variables, and transitions in ordinary language, and record the consolidation of the participation definition, the two-space articulation of local and aggregate description, and the forward articulations planted for subsequent development.

The central claim remains: entrepreneurial ecosystems emerge when entrepreneurial action streams, under sufficient structural conditions, become activated and then coherent in ways that reinforce the conditions of future entrepreneurial participation.